

Solution

Note: A similar problem is solved in XI[th] Physics volume I, unit 3 section 3.3.2.

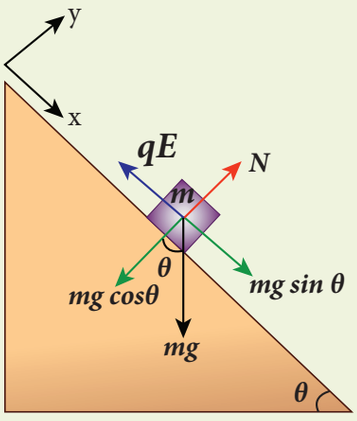
There are three forces that acts on the mass m:

(i) The downward gravitational force exerted by the Earth (mg)

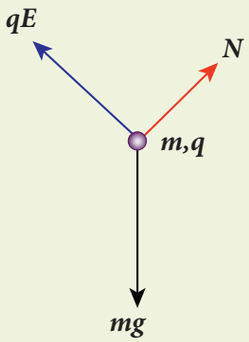
(ii) The normal force exerted by the inclined surface (N)

(iii) The Coulomb force given by uniform electric field (qE)

The free body diagram for the mass m is drawn below.



Forces acting on the mass *m*



Free body diagram

A convenient inertial coordinate system is located in the inclined surface as shown in the figure. The mass m has zero net acceleration both in x and y-direction.

Along x-direction, applying Newton's second law, we have

$$mg\sin\theta\hat{i} - qE\hat{i} = 0$$

$$mg\sin\theta - qE = 0$$

$$\text{or,} \quad E = \frac{mg\sin\theta}{q}$$

Note that the magnitude of the electric field is directly proportional to the mass m and inversely proportional to the charge q. It implies that, if the mass is increased by keeping the charge constant, then a strong electric field is required to stop the object from sliding. If the charge is increased by keeping the mass constant, then a weak electric field is sufficient to stop the mass from sliding down the plane.

The electric field also can be expressed in terms of height and the length of the inclined surface of the plane.

$$E = \frac{mg\,h}{qL}$$

1.3.4 Electric field lines

Electric field vectors are visualized by the concept of electric field lines. They form a set of continuous lines which are the visual representation of the electric field in some region of space. The following rules are followed while drawing electric field lines for charges.

- The electric field lines start from a positive charge and end at negative charges or at infinity. For a positive

UNIT-1(XII-Physics_Vol-1).indd 18 04-03-2019 10:25:09